CASO DE EJEMPLO

Below is an end-to-end worked example of a convergent mixed-methods study on “Perception of Safety and Waiting-Line Experiences” at a Lima Metropolitano station. First we invent realistic data (in text form), then show how you’d analyze the quantitative (survey) and qualitative (interviews) strands separately, and finally how you merge them into a unified proposal.

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1. Invented Data

1.1. Survey (Quantitative)

- **Sample size:** N = 150 regular commuters
- **Demographics (summary):**
  - 55 % female, 45 % male
  - Age: 18–25 (30 %), 26–40 (50 %), 41–60 (20 %)
- **Key Likert items (1 = Strongly Disagree … 5 = Strongly Agree):**
  - **Safety-1:** “I feel safe from crime while on the platform.”
  - **Safety-2:** “Lighting and visibility on the platform are adequate.”
  - **Wait-1:** “I find the queue at ticket machines to move quickly.”
  - **Wait-2:** “Signage guiding queues is clear and helpful.”
- **Example of 10 respondents** (each row is one commuter):

ID	Age	Gender	Freq. use/wk	Safety-1	Safety-2	Wait-1	Wait-2
C01	22	F	10	4	3	2	3
C07	35	M	15	3	2	1	2
C12	28	F	12	2	2	2	1
C23	45	M	7	5	4	4	4
C34	31	F	20	3	3	3	2
C45	39	M	14	2	1	2	2
C56	24	F	8	4	4	3	3
C67	52	M	5	5	5	5	5
C78	29	F	18	3	3	2	2
C89	33	M	11	2	2	1	1

1.2. Interviews (Qualitative)

- **Number of interviews:** 10
- **Participant codes:** P1, P2, …, P10 (gender and age noted)
- **Example excerpts:**
    1. **P1 (F, 24):** "At night the platform feels shadowy. I've seen people loitering near the turnstiles."
    2. **P4 (M, 37):** "Sometimes the ticket machine jams, and everyone rushes to the other machine – the line snakes around."
    3. **P6 (F, 31):** "The announcements about train arrivals are so quiet, I miss them and then the wait feels endless."
    4. **P9 (M, 47):** "Security guards are present but often chatting; I'd feel safer if they patrolled more."
    5. *(...plus 5 more quotes on lighting, crowding, machine usability, signage…)*

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2. Quantitative Data Analysis

Design: Convergent (both strands collected concurrently).

Steps in SPSS (or similar):

1. **Data Cleaning**
    - Check for missing values, out-of-range.
    - Replace or delete as per missing-data rules.
2. **Reliability Analysis**
    - Compute Cronbach's α for the "Safety" scale (Safety-1 & Safety-2) and for the "Waiting" scale (Wait-1 & Wait-2).
    - Accept if $\alpha \geq 0.70$.
3. **Descriptive Statistics**
    - Mean, SD for each item and scale.
    - Frequency distributions for demographic variables.
4. **Exploratory Factor Analysis (optional)**
    - Confirm that safety and waiting items load onto two distinct factors.
5. **Group Comparisons**
    - **t-tests or ANOVA:** Compare "perceived safety" by gender or age group.

    - **Chi-square:** Association between high/low safety perception (dichotomized) and frequency of use (e.g., ≤10 vs. >10 trips/week).
6. **Correlation & Regression**
    - Pearson's r between the Safety scale and Waiting scale.
    - Regression to predict overall satisfaction (if included) from safety and wait scales.

Example outcome:

- Cronbach's α: Safety = 0.82; Waiting = 0.78
- Mean Safety = 3.1 (SD 0.9); Mean Waiting = 2.5 (SD 1.0)
- Women report significantly lower safety (M=2.9) than men (M=3.3), $t(148)=2.4$, $p<0.05$.

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3. Qualitative Data Analysis

Here we follow **Braun & Clarke's six-phase thematic analysis**, illustrating each step with our invented interview excerpts.

3.1. Familiarization

- **Read transcripts** in full, jotting "first impressions."
- E.g., note recurrent mentions of poor lighting, idle staff, machine jams.

3.2. Initial (Open) Coding

- **Line-by-line**, assign codes to meaningful units.
- **Sample codes**:
    - "poor lighting"
    - "machine failure"
    - "guard inactivity"
    - "lack of signage"
    - "missed announcements"

Excerpt	Initial Code
"shadowy"	poor lighting
"machine jams"	machine failure
"guards chatting"	guard inactivity
"line snakes around"	chaotic queue

3.3. Generating Themes (Axial Coding)

- **Group related codes** into candidate themes:
    1. **“Physical Infrastructure Issues”**
        - poor lighting
        - lack of signage
    2. **“Operational Inefficiencies”**
        - machine failure
        - chaotic queue
        - missed announcements
    3. **“Perceived Security”**
        - guard inactivity
        - loitering passengers

3.4. Reviewing Themes

- **Check coherence**: Do excerpts within “Operational Inefficiencies” really belong together?
- **Revise**: perhaps split off “communication failures” (missed announcements) into its own minor theme.

3.5. Defining & Naming Themes

- Finalize theme definitions and ensure each is distinct:
    - **Infrastructure:** lighting, signage
    - **Equipment Reliability:** ticket machines, announcement systems
    - **Staff & Security Presence:** guards, crowd management

3.6. Producing the Report

- **Select vivid quotes** under each theme.
- E.g.,

> *“When that announcement crackles… I can’t tell if it’s my train or the next. I ended up waiting 15 more minutes.”* (P6)

4. Convergence & Unified Proposal

4.1. Bringing Strands Together

- **Joint Display Table** (example summary):

Theme	Qual (themes/quotes)	Quant (mean ± SD)
Lighting & Signage	"Platform feels shadowy…"	Safety-2: 2.8 ± 1.1
Machine & Announcements	"Machine jams… missed announcements"	Wait-1: 2.5 ± 1.0; Wait-2: 2.7 ± 1.0
Staff Presence	"Guards chatting…"	Safety-1: 3.1 ± 0.9

- **Interpretation:** Low scores on Safety-2 align with codes about poor lighting; moderate Waiting scores echo frustrations with machines and information flow.

4.2. Meta-Inferences & Proposal

- **Meta-Inference:** Infrastructure deficits and operational breakdowns jointly erode commuters' sense of safety and efficiency.
- **Recommendations:**
  1. **Improve Lighting & Signage**
     - Install LED overhead lights; add high-contrast queue markings.
  2. **Upgrade Ticketing & PA Systems**
     - Modernize ticket machines; ensure clear, loud announcements.
  3. **Enhance Staff Deployment**
     - Train guards for active patrols; station helpers to manage queues.

By weaving together the numerical trends (e.g., mean scores, gender differences) with the rich, contextual insights from interviews, your thesis can offer a deeply grounded, actionable plan—demonstrating the true power of mixed methods in social-scientific research.

You said:

explain the p value with my example to a noob in statistics

ChatGPT said:

Think of the p-value as a "surprise meter" for your data—how surprised you'd be to see a difference at least as big as the one you found, **if in reality there were no difference at all** between groups.

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1. Set up the "no-difference" scenario (the null hypothesis)

In our Lima Metropolitano example, we compared the average safety rating for women ($M = 2.9$) versus men ($M = 3.3$).

- **Null hypothesis (H_0):** Women and men have the **same** true average safety rating.
- **Alternative hypothesis (H_1):** They have **different** true averages.

2. Compute the p-value

Using a t-test, you calculate a statistic that measures how far apart the two sample means are, in units of their combined variability. From that t-statistic you get a p-value. This p-value answers:

"If H_0 were true (no real difference), what is the probability of seeing a difference between women's and men's safety ratings as large as—or larger than—the one we actually observed?"

3. Interpret the p-value

- If the p-value is **large** (say, 0.30), that means "there's a 30% chance of seeing this kind of difference just by random luck." That's not very surprising—so you **fail to reject** the no-difference idea.
- If the p-value is **small** (say, 0.02), that means "there's only a 2% chance of getting a difference this big if there really were no difference." That is quite surprising—so you **reject** the no-difference idea and conclude there probably **is** a real difference.

In our write-up we had

$t(148) = 2.4, p < 0.05$

This means:

- Our test statistic was 2.4.
- The calculated p-value was somewhere below 0.05 (often you'll see something like 0.02 or 0.03).
- Because $p < 0.05$ (a common cutoff), we decided that the observed gap (3.3 vs. 2.9) is **unlikely** to be due to chance alone.

4. Putting it in plain English

"If women and men really felt equally safe on the platform, there's less than a 5% chance we'd see a difference in their average safety ratings as large as we did in our survey. Since that's pretty unlikely, we conclude that women really do feel less safe than men in this station."

Key take-aways for a statistics noob

1. **p-value ≠ probability that H_0 is true.**
2. **p-value = probability of your data (or more extreme) if H_0 is true.**
3. A **small p-value** (commonly < 0.05) means "data are surprising under H_0," so you reject H_0.
4. A **large p-value** means "data aren't surprising under H_0," so you don't have evidence to say there's a real effect.

With that intuition, whenever you report "$p < 0.05$," you're really saying "what I saw in my data would be pretty rare—less than 5% likely—if nothing were actually going on." That's why we then believe something *is* going on.